

Question And Answer

Manual Testing Assignment 2– QA

Product Context

You are testing a **Web-based Food Delivery Application**

The application allows users to:

- Register and login
 - Search restaurants
 - Add food items to cart
 - Apply coupons
 - Place orders
 - Track order status
-

Part 1 – Bug Reporting (Realistic Scenario)

Below are issues observed during testing.

Scenario 1

User adds items worth ₹499 to the cart.

A coupon **SAVE50** is applicable only on **orders above ₹500**.

When the user applies the coupon:

- The discount **still gets applied**.
- Final price becomes ₹449.

Task

Write a **complete bug report** including:

- Bug ID
- Title
- Environment
- Steps to reproduce
- Expected Result
- Actual Result
- Severity
- Priority

- Attachments (if any)

Answer



Bug Report – Scenario 1

Bug ID: BUG-001

Title: Coupon SAVE50 applied for order value below minimum requirement

Environment:

- Application: Web-based Food Delivery Application
- Browser: Google Chrome (latest version)
- OS: Windows 10 / macOS

Steps to Reproduce:

1. Open the food delivery application
2. Login with valid credentials
3. Add food items worth ₹499 to the cart
4. Apply coupon code SAVE50
5. Observe the final price

Expected Result:

Coupon SAVE50 should NOT be applied since the minimum order value requirement (₹500) is not met.

Actual Result:

Coupon is successfully applied even though the order value is ₹499, and the final price becomes ₹449.

Severity: Medium

(Incorrect pricing affects business logic but does not crash the system)

Priority: High

(Direct impact on revenue and pricing rules, needs quick fix)

Attachments (if any):

- Screenshot of cart before applying coupon
- Screenshot showing applied discount and final price

Scenario 2

When a user logs out and presses **browser back button**, the **dashboard page appears again** and user can access order history.

Task

Report this as a **security bug**.

Part 2 – Black Box Testing

Test the **Login Page**.

Login Page Fields

- Email
- Password
- Login Button

Requirements

Email rules:

- Must contain @
- Must contain domain (.com, .in, etc)

Password rules:

- Minimum **8 characters**
 - Must contain **1 uppercase**
 - Must contain **1 number**
-

Answer



Part 1 – Scenario 2 (Security Bug Report)

Bug ID: BUG-002

Title: Unauthorized access to dashboard after logout using browser back button

Environment:

- Application: Web-based Food Delivery Application
- Browser: Google Chrome (latest version)
- OS: Windows/macOS

Steps to Reproduce:

1. Login with valid credentials
2. Navigate to dashboard
3. Click on Logout
4. Press the browser Back button
5. Observe the page

Expected Result:

User should be redirected to the login page or session should be expired. Dashboard access must be restricted after logout.

Actual Result:

Dashboard page is displayed and user can access order history even after logout.

Severity: Critical 🚨

(Security vulnerability — unauthorized access to user data)

Priority: High

(Needs immediate fix due to data exposure risk)

Part 2 – Black Box Testing (Login Page)

👉 We use:

- **Equivalence Partitioning (EP)** → Valid vs Invalid inputs
- **Boundary Value Analysis (BVA)** → Edge cases (like password length = 8)

✅ Test Cases

Test Case ID	Scenario	Input	Expected Result
TC01	Valid login	valid@mail.com / Pass1234	Login successful
TC02	Email without @	testmail.com	Error message
TC03	Email without domain	test@	Error message
TC04	Password less than 8 chars (BVA)	Pass12	Error message
TC05	Password exactly 8 chars (BVA)	Pass1234	Login successful
TC06	No uppercase in password	pass1234	Error message
TC07	No number in password	Password	Error message
TC08	Empty email & password	""	Error message

Task

1. Write **test cases using Black Box techniques**
2. Use:
 - **Equivalence Partitioning**
 - **Boundary Value Analysis**
3. Provide **at least 8 test cases**

Example format:

Test Case ID	Scenario	Input	Expected Result
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Part 3 – White Box Testing (Logic Based)

Consider the following pseudo code for applying a discount.

```
if (orderAmount > 500)
{
    if (userType == "PREMIUM")
        discount = 20%
    else
        discount = 10%
}
else
{
    discount = 0
}
```

Answer



Part 3 – White Box Testing (Logic Based)

◇ Given Logic

```
if (orderAmount > 500)
{
    if (userType == "PREMIUM")
        discount = 20%
    else
        discount = 10%
}
else
{
    discount = 0
}
```



1. Decision Table

Condition / Case	Case 1	Case 2	Case 3
orderAmount > 500	Yes	Yes	No
userType = PREMIUM	Yes	No	-
Discount	20%	10%	0%



2. Test Cases (All Branch Coverage)

Test Case ID	orderAmount	userType	Expected Result
TC01	600	PREMIUM	20% discount
TC02	600	NORMAL	10% discount
TC03	400	PREMIUM/NORMAL	0% discount



3. Cyclomatic Complexity

👉 Count decision points:

- if (orderAmount > 500)

- `if (userType == "PREMIUM")`

👉 Formula:

Cyclomatic Complexity = Number of decision points + 1

👉 So:

$M = 2 + 1 = 3$

Tasks

1. Draw the **Decision Table**
2. Identify **test cases to cover all branches**
3. Calculate **Cyclomatic Complexity**